

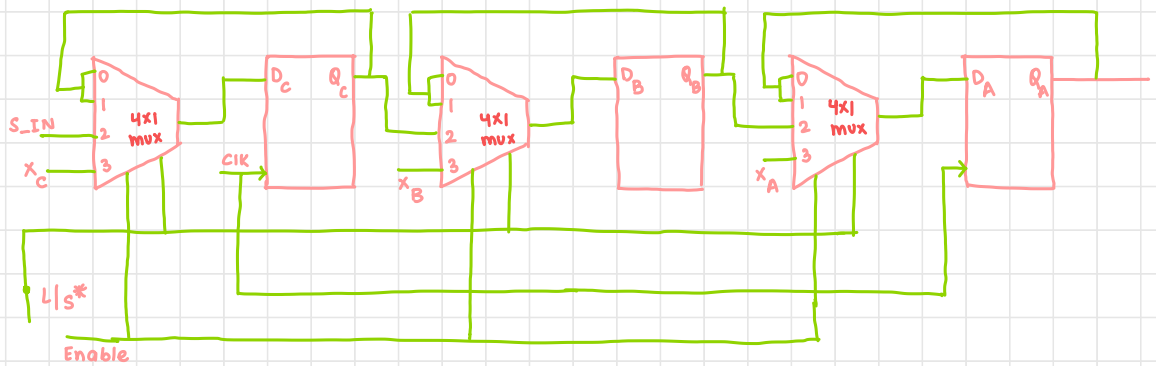
Problem Set 6

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1. Observing the truth table

EN	L/S^*	D_C	D_B	D_A	Function
0	0	Q_C	Q_B	Q_A	Hold
0	1	Q_C	Q_B	Q_A	Hold
1	0	S_IN	Q_C	Q_B	Shift
1	1	C	B	A	Load

Drawing the circuit:



2.a To implement a gated binary counter, we can use a JK flip-flop with a gate input that enables or disables the counter. The circuit below shows a 3-bit binary counter with a gated enable input.

Present Next

Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	J_2	K_2	J_1	K_1	J_0	K_0
0	0	0	0	0	1	1	0	x	0	1	x
0	0	1	0	1	0	0	x	1	x	x	1
0	1	0	0	1	1	0	x	x	0	1	x
0	1	1	1	0	0	1	x	x	1	x	1
1	0	0	1	0	1	x	0	0	x	1	x
1	0	1	1	1	0	x	0	1	x	x	1
1	1	0	1	1	1	x	0	x	0	1	x
1	1	1	0	0	0	x	1	x	1	x	1

Q_2, Q_0

$Q_2 \backslash Q_0$	00	01	11	10
0	0	0	1	0
1	x	x	x	x

K-map for J_2

Q_2, Q_0

$Q_2 \backslash Q_0$	00	01	11	10
0	x	x	x	x
1			1	

K-map for K_2

$$J_2 = K_2 = Q_1 Q_0$$

Q_1, Q_0

$Q_1 \backslash Q_0$	00	01	11	10
0	0	1	x	x
1	0	1	x	x

K-map for J_1

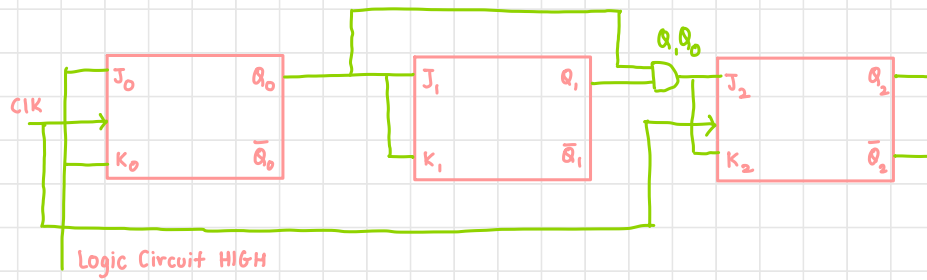
Q_1, Q_0

$Q_1 \backslash Q_0$	00	01	11	10
0	x	x	1	0
1	x	x	1	0

K-map for K_1

$$J_1 = K_1 = Q_0$$

$$J_0 = 1 = K_0$$



b.

Q_2	Q_1	Q_0	P	$\bar{Q}_2 = Q_2$	$\bar{Q}_1 = Q_1$	$\bar{Q}_0 = Q_0$
0	0	0	0	0	0	1
0	0	1	0	0	1	0
0	1	0	1	0	1	1
0	1	1	1	1	0	0
1	0	0	0	1	0	1
1	0	1	1	1	1	0
1	1	0	0	1	1	1
1	1	1	1	0	0	0

Q_2, Q_0

$Q_2 \backslash Q_0$	00	01	11	10
0			1	1
1			1	1

$$P = \bar{Q}_2 Q_1 + Q_2 Q_0$$

Q_1, Q_0

$Q_1 \backslash Q_0$	00	01	11	10
0		1		1
1		1		1

$$D_1 = \bar{Q}_1 Q_0 + Q_1 \bar{Q}_0 = Q_1 \oplus Q_0$$

Q_2, Q_0

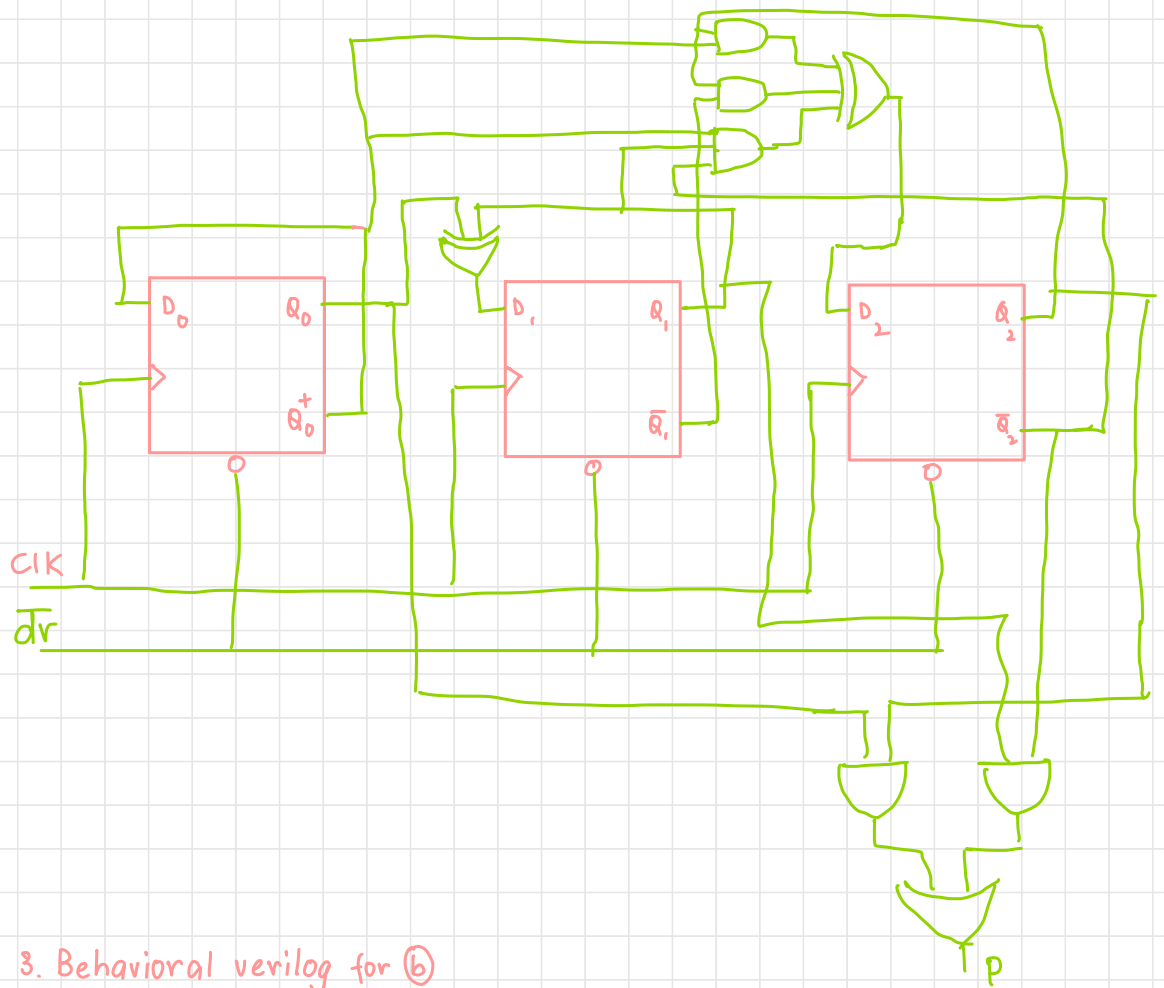
$Q_2 \backslash Q_0$	00	01	11	10
0			1	
1	1	1	1	1

$$D_2 = Q_2 \bar{Q}_0 + Q_2 \bar{Q}_1 + \bar{Q}_2 Q_1 Q_0$$

Q_1, Q_0

$Q_1 \backslash Q_0$	00	01	11	10
0	1	1	1	1
1	1	1	1	1

$$D_0 = \bar{Q}_0$$



3. Behavioral verilog for (b)

```

1 module dff_pulse(dk,reset,p,d);
2   input register dk;
3   output register p;
4   reg[0:2]d;
5   always@(posedge dk or posedge reset)
6     if(reset == 0) begin
7       p<=0;
8       q<=0;
9     end
10  else begin
11    d<=d + 3'b001;
12    if(d==3'b010 or d==3'b011 or d==3'b101 or d==3'b111) begin
13      p<=1;
14    end
15    else begin
16      p<=0;
17    end
18  end
19 endmodule
20

```

SV/Veril

3. // Behavioral Verilog for @

```

1 module binary_counter(clk,E,a,b,c)
2     input clk;
3     input E;
4     output res[0:2] q;
5
6     always@(posedge clk)
7     begin
8         if (E) begin
9             if (out == 3'b111) begin
10                 q <= 3'b000;
11             end
12             else begin
13                 q <= q+1;
14             end
15         end
16     end
17 endmodule
18

```

4.

For Synchronous modulo-14 counter

Present State				Next State				J1	K1	J2	K2	J3	K3	J4	K4
Q1	Q2	Q3	Q4	Q1'	Q2'	Q3'	Q4'								
0	0	0	0	0	0	0	1	0	x	0	x	0	x	1	x
0	0	0	1	0	0	1	0	0	x	0	x	1	x	x	1
0	0	1	0	0	0	1	1	0	x	0	x	x	0	1	x
0	0	1	1	0	1	0	0	0	x	1	x	x	1	x	1
0	1	0	0	0	1	0	1	0	x	x	0	0	x	1	x
0	1	0	1	0	1	1	0	0	x	x	0	1	x	x	1
0	1	1	0	0	1	1	1	0	x	x	0	x	0	1	x
0	1	1	1	1	1	0	0	1	x	x	1	x	1	x	1
1	0	0	0	1	0	0	1	x	0	0	x	0	x	1	x
1	0	0	1	1	0	1	0	x	0	0	x	1	x	x	1
1	0	1	0	1	0	1	1	x	0	0	x	x	0	1	x
1	0	1	1	1	1	0	0	x	0	1	x	x	1	x	1
1	1	0	0	1	1	0	1	x	0	x	0	0	x	1	x
1	1	0	1	1	1	1	0	x	0	x	0	1	x	x	1
1	1	1	0	1	1	1	1	x	0	x	0	x	0	1	x
1	1	1	1	0	1	1	1	x	0	x	0	x	0	1	x
1	1	1	1	1	0	0	0	x	1	x	1	x	1	x	1

J_1, Q_1, Q_2, Q_3, Q_4

	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	x	x	x	x
10	x	x	x	x

K_1, Q_1, Q_2, Q_3, Q_4

Q_1	Q_2	Q_3	Q_4
00	01	11	10
00	x	x	x
01	x	x	x
11	0	0	1
10	0	0	0

J_2, Q_1, Q_2, Q_3, Q_4

Q_1	Q_2	Q_3	Q_4	
00	0	0	1	0
01	x	x	x	x
11	x	x	x	x
10	0	0	1	0

K_2, Q_1, Q_2, Q_3, Q_4

Q_1	Q_2	Q_3	Q_4	
00	x	x	x	x
01	0	0	1	0
11	0	0	1	0
10	x	x	x	x

J_3, Q_1, Q_2, Q_3, Q_4

$Q_1 Q_2$	00	01	11	10
00	0	1	x	x
01	0	1	x	x
11	0	1	x	x
10	0	1	x	x

K_3, Q_1, Q_2, Q_3, Q_4

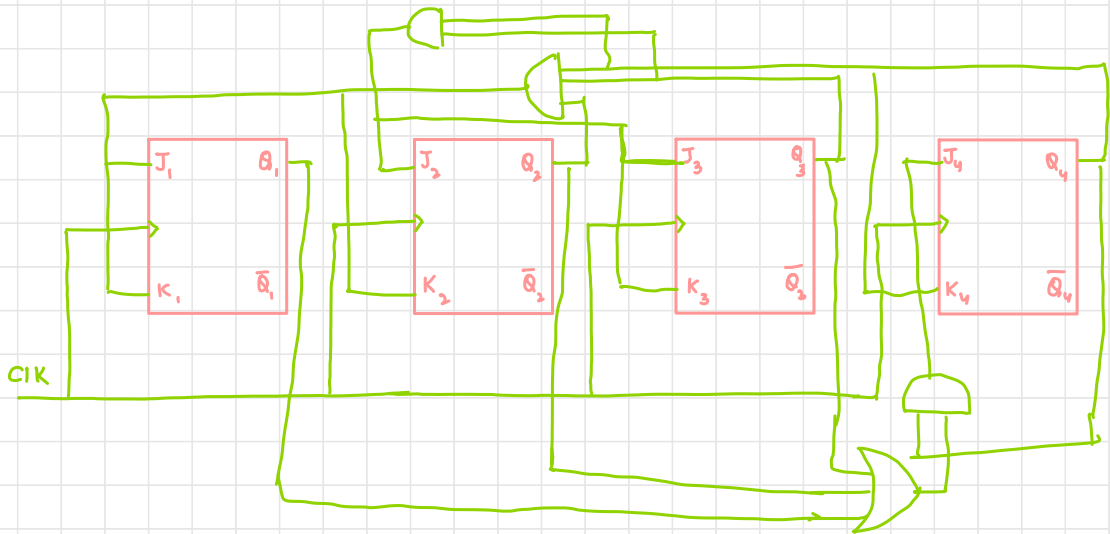
Q_1	Q_2	Q_3	Q_4
00	x	x	1
01	x	x	1
11	x	x	1
10	x	x	1

J_4, Q_1, Q_2, Q_3, Q_4

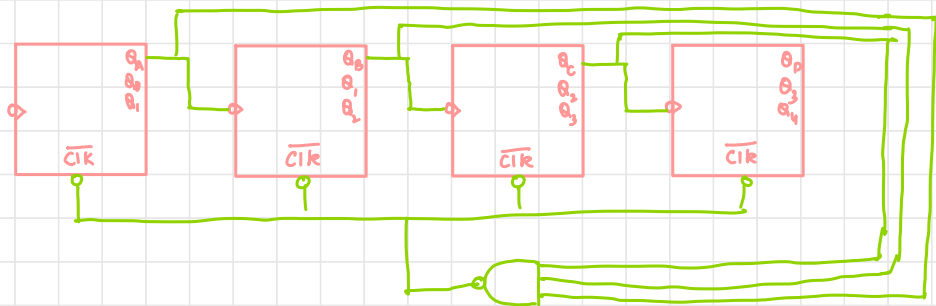
Q1	Q2	Q3	Q4	
00	1	x	x	1
01	1	x	x	0
11	1	x	x	1
10	1	x	x	1

K_4, Q_1, Q_2, Q_3, Q_4

	Q_1	Q_2	Q_3	Q_4
Q_4	00	01	11	10
00	x	1	1	x
01	x	1	1	x
11	x	1	1	x
10	x	1	1	x



Asynchronous modulo-14 counter



5.

For Synchronous modulo-15 counter

Present State				Next State							
Q3	Q2	Q1	Q0	Q3'	Q2'	Q1'	Q0'	D3	D2	D1	D0
0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	0	1	0	0	0	1	0
0	0	1	0	0	0	1	1	0	0	1	1
0	0	1	1	0	1	0	0	0	1	0	0
0	1	0	0	0	1	0	1	0	1	0	1
0	1	0	1	0	1	1	0	0	1	1	0
0	1	1	0	0	1	1	1	0	1	1	1
0	1	1	1	1	0	0	0	1	0	0	0
1	0	0	0	1	0	0	1	0	0	1	1
1	0	0	1	1	0	1	0	0	1	0	0
1	0	1	0	1	0	1	1	0	1	1	1
1	0	1	1	1	1	0	0	1	0	0	0
1	1	0	0	1	1	0	1	1	0	1	1
1	1	0	1	1	1	1	0	1	1	0	0
1	1	1	0	1	1	1	1	1	1	1	1
1	1	1	1	0	0	0	0	0	0	0	0

K-map for D_3

Q_3, Q_2 \ Q_1, Q_0	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	1	1	0	1
10	1	1	1	1

$$D_3 = Q_3 \oplus Q_2 Q_1 Q_0$$

K-map for D_2

Q_3, Q_2 \ Q_1, Q_0	00	01	11	10
00	0	0	1	0
01	1	1	0	1
11	1	1	0	1
10	0	0	1	0

$$D_2 = Q_2 \oplus (Q_1 Q_0)$$

K-map for D_1

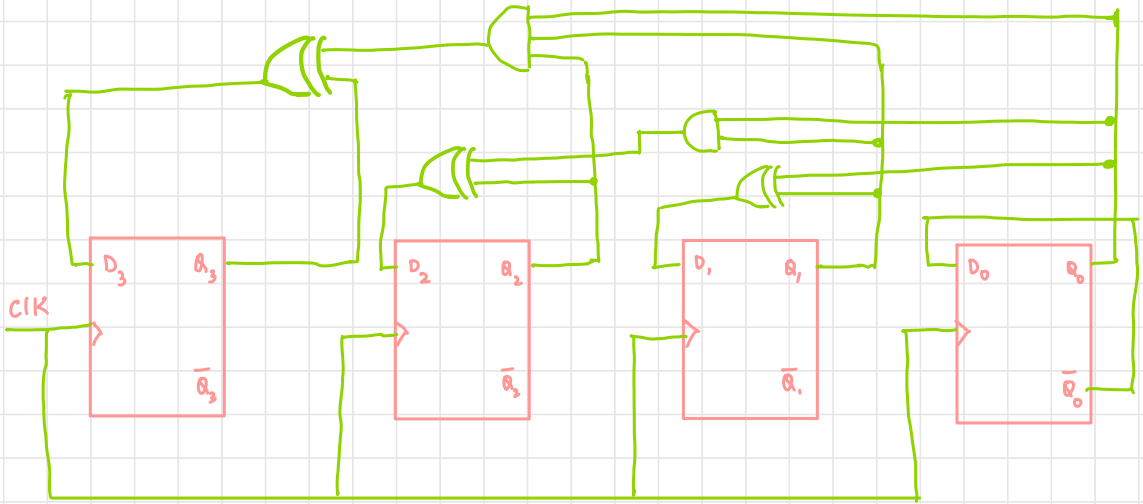
Q_3, Q_2 \ Q_1, Q_0	00	01	11	10
00	0	1	0	1
01	0	1	0	1
11	0	1	0	1
10	0	1	0	1

$$D_1 = Q_1 \oplus Q_0$$

K-map for D_0

Q_3, Q_2 \ Q_1, Q_0	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	1	0	0	1
10	1	0	0	1

$$D_0 = \bar{Q}_0$$



6. The expression for f_1 is :

$$\begin{aligned} \bar{f}_1 &= \sum m(2, 9, 17, 30, 60) \\ &= \bar{F}\bar{E}\bar{D}\bar{C}\bar{B}\bar{A} + \bar{F}\bar{E}\bar{D}\bar{C}\bar{B}A + \bar{F}\bar{E}\bar{D}\bar{C}B\bar{A} + \bar{F}\bar{E}\bar{D}C\bar{B}\bar{A} + \bar{F}\bar{E}DC\bar{B}\bar{A} \end{aligned}$$

The expression for f_2 is .

$$\begin{aligned} \bar{f}_1 &= \sum m(2, 8, 15, 35, 56) \\ &= \bar{F}\bar{E}\bar{D}\bar{C}\bar{B}\bar{A} + \bar{F}\bar{E}\bar{D}\bar{C}\bar{B}A + \bar{F}\bar{E}\bar{D}C\bar{B}\bar{A} + \bar{F}\bar{E}\bar{D}C\bar{B}A + \bar{F}\bar{E}DC\bar{B}\bar{A} \end{aligned}$$

The expression for f_3 is .

$$\begin{aligned}\bar{f}_3 &= \sum m(1, 8, 16, 37, 63) \\ &= \bar{F}\bar{E}\bar{D}\bar{C}\bar{B}\bar{A} + \bar{F}\bar{E}\bar{D}\bar{C}\bar{B}\bar{A} + \bar{F}\bar{E}\bar{D}\bar{C}\bar{B}\bar{A} + \bar{F}\bar{E}\bar{D}\bar{C}\bar{B}\bar{A} + \bar{F}\bar{E}\bar{D}\bar{C}\bar{B}\bar{A}\end{aligned}$$

The expression for f is .

$$\begin{aligned}\bar{f} &= \sum m(3, 27, 39, 41, 63) \\ &= \bar{F}\bar{E}\bar{D}\bar{C}\bar{B}\bar{A} + \bar{F}\bar{E}\bar{D}\bar{C}\bar{B}\bar{A} + \bar{F}\bar{E}\bar{D}\bar{C}\bar{B}\bar{A} + \bar{F}\bar{E}\bar{D}\bar{C}\bar{B}\bar{A} + \bar{F}\bar{E}\bar{D}\bar{C}\bar{B}\bar{A}\end{aligned}$$

